KAIZEN MATHS

Lesson starter

Volumes of Revolution

y-axis and Parameter Problems

Name _____

S1

Rotate about the y-axis by integrating with respect to y.

The region under $x = 3y + 2$ from $y = 0$ to $y = 2$ is rotated about the y-axis. Find the volume.

S2

Rotate about the y-axis by integrating with respect to y.

The region under $x = \sqrt{2y + 1}$ from $y = 0$ to $y = 2$ is rotated about the y-axis. Find the volume.

S3

Use the given volume to find the unknown parameter.

The region under $y = kx$ from $x = 0$ to $x = 2$ is rotated about the x-axis. The volume is $\frac{32\pi}{3}$. Find k, where $k > 0$.

S4

Rotate about the y-axis by integrating with respect to y.

The region under $x = 2y + 1$ from $y = 0$ to $y = 2$ is rotated about the y-axis. Find the volume.

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S4

Rotate about the y-axis by integrating with respect to y.

The region under $x = 2y + 1$ from $y = 0$ to $y = 2$ is rotated about the y-axis. Find the volume.

ALG-3EBB-L4 · S

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KAIZEN MATHS

Exit ticket

Volumes of Revolution

y-axis and Parameter Problems

Name

EA1

Rotate about the y-axis by integrating with respect to y.

The region under $x = \sqrt{3y + 3}$ from $y = 0$ to $y = 2$ is rotated about the y-axis. Find the volume.

EA2

Use the given volume to find the unknown parameter.

The region under $y = kx$ from $x = 0$ to $x = 3$ is rotated about the x-axis. The volume is 81π. Find k, where $k > 0$.

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KAIZEN MATHS

Exit ticket

Volumes of Revolution

y-axis and Parameter Problems

Name

EB1

Rotate about the y-axis by integrating with respect to y.

The region under $x = y + 1$ from $y = 0$ to $y = 3$ is rotated about the y-axis. Find the volume.

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Volumes of Revolution

y-axis and Parameter Problems

Name

EA1

Rotate about the y-axis by integrating with respect to y.

The region under $x = \sqrt{3y + 3}$ from $y = 0$ to $y = 2$ is rotated about the y-axis. Find the volume.

EA2

Use the given volume to find the unknown parameter.

The region under $y = kx$ from $x = 0$ to $x = 3$ is rotated about the x-axis. The volume is 81π. Find k, where $k > 0$.

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KAIZEN MATHS

Exit ticket

Volumes of Revolution

y-axis and Parameter Problems

Name

EB1

Rotate about the y-axis by integrating with respect to y.

The region under $x = y + 1$ from $y = 0$ to $y = 3$ is rotated about the y-axis. Find the volume.

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Teacher answers & working

Volumes of Revolution · y-axis and Parameter Problems

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S1

Rotate about the y-axis by integrating with respect to y.

The region under $x = 3y + 2$ from $y = 0$ to $y = 2$ is rotated about the y-axis. Find the volume.

Answer:

$V = 56\pi$

Working:

For rotation about the y-axis in this form, use $V = \pi \int_c^d x^2 \, dy$.

Here $x = 3y + 2$ and the bounds are $y = 0$ to $y = 2$.

Set up: $V = \pi \int_0^2 (3y + 2)^2 \, dy$.

Square the radius: $(3y + 2)^2 = 9y^2 + 12y + 4$.

Integrate with respect to y: $\int (9y^2 + 12y + 4) \, dy = 3y^3 + 6y^2 + 4y$.

Evaluate: $\left[3y^3 + 6y^2 + 4y\right]_0^2 = 56$.

Therefore $V = 56\pi$.

S2

Rotate about the y-axis by integrating with respect to y.

The region under $x = \sqrt{2y + 1}$ from $y = 0$ to $y = 2$ is rotated about the y-axis. Find the volume.

Answer:

$V = 6\pi$

Working:

For rotation about the y-axis, use $V = \pi \int_c^d x^2 \, dy$.

Here $x = \sqrt{2y + 1}$, so $x^2 = 2y + 1$.

Set up: $V = \pi \int_0^2 (2y + 1) \, dy$.

Integrate with respect to y: $\int (2y + 1) \, dy = y^2 + y$.

Evaluate: $\left[y^2 + y\right]_0^2 = 6$.

Therefore $V = 6\pi$.

Teacher answers & working

Volumes of Revolution · y-axis and Parameter Problems

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S3

Use the given volume to find the unknown parameter.

The region under $y = kx$ from $x = 0$ to $x = 2$ is rotated about the x-axis. The volume is $\frac{32\pi}{3}$. Find k, where $k > 0$.

Answer:

$k = 2$

Working:

For rotation about the x-axis, use $V = \pi \int_a^b y^2 \, dx$.

Here $y = kx$, so $y^2 = k^2x^2$.

Set up the volume: $V = \pi \int_0^2 k^2x^2 \, dx$.

Treat k^2 as a constant: $V = \pi k^2 \int_0^2 x^2 \, dx$.

Integrate and substitute the bounds: $\int_0^2 x^2 \, dx = \left[\frac{x^3}{3}\right]_0^2 = \frac{8}{3}$.

So $V = \frac{8}{3}k^2\pi$.

Compare with the given volume: $\frac{8}{3}k^2\pi = \frac{32\pi}{3}$.

Cancel π and solve: $k^2 = 4$, so $k = 2$ because $k > 0$.

S4

Rotate about the y-axis by integrating with respect to y.

The region under $x = 2y + 1$ from $y = 0$ to $y = 2$ is rotated about the y-axis. Find the volume.

Answer:

$V = \frac{62\pi}{3}$

Working:

For rotation about the y-axis in this form, use $V = \pi \int_c^d x^2 \, dy$.

Here $x = 2y + 1$ and the bounds are $y = 0$ to $y = 2$.

Set up: $V = \pi \int_0^2 (2y + 1)^2 \, dy$.

Square the radius: $(2y + 1)^2 = 4y^2 + 4y + 1$.

Integrate with respect to y: $\int (4y^2 + 4y + 1) \, dy = \frac{4}{3}y^3 + 2y^2 + y$.

Evaluate: $\left[\frac{4}{3}y^3 + 2y^2 + y\right]_0^2 = \frac{62}{3}$.

Therefore $V = \frac{62\pi}{3}$.

Teacher answers & working

Volumes of Revolution · y-axis and Parameter Problems

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EA1

Rotate about the y-axis by integrating with respect to y.

The region under $x = \sqrt{3y+3}$ from $y = 0$ to $y = 2$ is rotated about the y-axis. Find the volume.

Answer:

$V = 12\pi$

Working:

For rotation about the y-axis, use $V = \pi \int_c^d x^2\,dy$.

Here $x = \sqrt{3y+3}$, so $x^2 = 3y+3$.

Set up: $V = \pi \int_0^2 (3y+3)\,dy$.

Integrate with respect to y: $\int (3y+3)\,dy = \frac{3}{2}y^2 + 3y$.

Evaluate: $\left[\frac{3}{2}y^2 + 3y\right]_0^2 = 12$.

Therefore $V = 12\pi$.

EA2

Use the given volume to find the unknown parameter.

The region under $y = kx$ from $x = 0$ to $x = 3$ is rotated about the x-axis. The volume is 81π . Find k , where $k > 0$.

Answer:

$k = 3$

Working:

For rotation about the x-axis, use $V = \pi \int_a^b y^2\,dx$.

Here $y = kx$, so $y^2 = k^2x^2$.

Set up the volume: $V = \pi \int_0^3 k^2x^2\,dx$.

Treat k^2 as a constant: $V = \pi k^2 \int_0^3 x^2\,dx$.

Integrate and substitute the bounds: $\int_0^3 x^2\,dx = \left[\frac{x^3}{3}\right]_0^3 = 9$.

So $V = 9k^2\pi$.

Compare with the given volume: $9k^2\pi = 81\pi$.

Cancel π and solve: $k^2 = 9$, so $k = 3$ because $k > 0$.

Teacher answers & working

Volumes of Revolution · y-axis and Parameter Problems

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EB1

Rotate about the y-axis by integrating with respect to y.

The region under $x = y + 1$ from $y = 0$ to $y = 3$ is rotated about the y-axis. Find the volume.

Answer:

$V = 21\pi$

Working:

For rotation about the y-axis in this form, use $V = \pi \int_c^d x^2 \, dy$.

Here $x = y + 1$ and the bounds are $y = 0$ to $y = 3$.

Set up: $V = \pi \int_0^3 (y+1)^2 \, dy$.

Square the radius: $(y+1)^2 = y^2 + 2y + 1$.

Integrate with respect to y: $\int (y^2 + 2y + 1) \, dy = \frac{1}{3}y^3 + y^2 + y$.

Evaluate: $\left[\frac{1}{3}y^3 + y^2 + y\right]_0^3 = 21$.

Therefore $V = 21\pi$.